

Learn HTML: Beginner's Guide

Lesson 6: Embedding Media

Embedding Media - Study Notes

Key Concepts

1. **HTML5 media elements** – `<audio>` and `<video>` provide native support for playing sound and video without plug ins.
2. **Source handling** – Use `<source>` elements (or the `src` attribute) to supply multiple file formats for broader browser compatibility.
3. **Controls & autoplay** – The `controls`, `autoplay`, `loop`, `muted`, and `preload` attributes control playback UI and behavior.
4. **Embedding external content** – `<iframe>` lets you embed another HTML page, maps, YouTube videos, PDFs, etc., inside your page.
5. **Accessibility & fallback** – Provide text alternatives (`<track>` for subtitles, fallback content inside the tags) to make media usable for all users.

Summary

HTML5 introduced dedicated tags for embedding media directly into a web page. The `<audio>` element streams sound files, while `<video>` does the same for video files. Both tags accept one or more `<source>` children, allowing you to list different file formats (e.g., MP3, OGG for audio; MP4, WebM, OGG for video) so the browser can pick the first it supports. Adding the `controls` attribute automatically generates a play pause bar, volume slider, and full screen button for video, eliminating the need for third party plugins.

The `<iframe>` tag creates an inline frame that displays another HTML document inside the current page. It is commonly used to embed YouTube videos, Google Maps, PDFs, or any external web content. Like media elements, iframes have attributes that affect their behavior—`src`, `width`, `height`, `title`, `allowfullscreen`, and sandboxing options for security. Proper use of these tags, combined with fallback content and accessibility features, ensures a smooth, inclusive experience for all users.

Important Points

- **Multiple sources:** `<source src="file.mp4" type="video/mp4">` !' browser picks the first supported format.
- **Essential attributes:**
 - `controls` – shows built in UI.
 - `autoplay` – starts playback automatically (often needs `muted` to work on mobile).
 - `loop` – repeats media.
 - `muted` – silences audio (required for autoplay on many browsers).
 - `preload` – hints how much data to load (`none`, `metadata`, `auto`).
- **Accessibility:**

- Use `<track kind="subtitles" src="sub.vtt" srclang="en" label="English">` for captions.
- Provide meaningful fallback text inside `<audio>` / `<video>` tags for browsers that don't support them.
 - **Iframe security:**
 - `sandbox` attribute can restrict scripts, forms, etc.
 - `allow` attribute defines what features (e.g., `camera`; `microphone`; `fullscreen`) are permitted.
 - **Responsive sizing:** Use CSS (`max-width: 100%; height: auto;`) to make media adapt to different screen sizes.

Examples

1. Audio with multiple formats and controls

```

<<<html
<audio controls preload="metadata">
  <source src="song.mp3" type="audio/mpeg">
  <source src="song.ogg" type="audio/ogg">
  <!-- Fallback text for very old browsers -->
  Your browser does not support the audio element.
</audio>
<<<

```

***Explanation:** The browser tries the MP3 first; if unsupported, it falls back to OGG. The `controls` attribute adds a play bar, and `preload="metadata"` loads only the file's header to show duration quickly.

2. Responsive video embedded from YouTube via iframe

```

<<<html
<div style="position:relative; padding-bottom:56.25%; height:0; overflow:hidden;">
  <iframe
    src="https://www.youtube.com/embed/dQw4w9WgXcQ"
    title="YouTube video player"
    style="position:absolute; top:0; left:0; width:100%; height:100%; border:none;"
    allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
    allowfullscreen>
  </iframe>
</div>
<<<

```

***Explanation:** The wrapper `div` creates a 16:9 aspect ratio box that scales with the viewport, making the video responsive. The `allowfullscreen` attribute lets users expand the video, and `allow` lists permitted features.

Quick Review

1. **What attribute adds a built in play/pause UI to `<audio>` and `<video>`?**
2. **Why should you provide more than one `<source>` format for a media file?**
3. **Name two security related attributes you can use with `<iframe>`.**

4. **How can you make an embedded video resize proportionally on different screen sizes?**
5. **What tag is used to add subtitles or captions to a `<video>` element?**

Further Reading

- **HTML Media Elements Specification** – deeper dive into attributes, events, and media APIs.
- **Responsive Media Techniques** – using `srcset``, `<picture>`, and CSS tricks for fluid layouts.
- **Web Accessibility Guidelines (WCAG) for Audio/Video** – ensuring captions, transcripts, and keyboard control.
- **Iframe Sandbox & CSP** – advanced security measures for embedding third party content.
- **JavaScript Media API** – controlling playback programmatically (play, pause, seek, volume).

